Fiscal Year:	FY 2018	**Task Last Updated:**	FY 10/01/2018
PI Name:	Loucas, Bradford Ph.D.		
Project Title:	The Contribution of Delta Rays to the Chromosome Aberration Dose Response in Human Cells Irradiated with HZE Particles of Different Energy but the Same LET		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	avondeak@nasaprs.com	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	409-772-9745
Organization Name:	University of Texas Medical Branch, Galveston		
PI Address 1:	Radiation Oncology		
PI Address 2:	301 University Blvd, 0884		
PI Web Page:			
City:	Galveston	**State:**	TX
Zip Code:	77555-5302	**Congressional District:**	14
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2013 Space Radiobiology NNJ13ZSA001N
Start Date:	12/20/2013	**End Date:**	05/19/2018
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Simonsen, Lisa	**Contact Phone:**	
Contact Email:	lisa.c.simonsen@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 5/19/2018 per S. Monk/HRP-LaRC (Ed., 12/21/17)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Cornforth, Michael Ph.D. (University of Texas Medical Branch, Galveston)		
Grant/Contract No.:	NNX14AC76G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Highly energetic heavy ions (HZE), of the type found in galactic cosmic rays, deposit energy in the form of ionizations along the tracks they take as they travel through a cell. These ionizations are tightly spaced along these tracks and this spacing makes them highly damaging to cellular structures. Some of the particles' energy is also imparted to electrons (delta rays) which can travel great distances and produce damage, albeit less severe, in cells far removed from the original particle track. The ranges of these electrons are proportional to the energy of the original HZE particle. While these delta rays by themselves are not likely to produce much in the way of measurable damage, we have evidence that these electrons can add damage to that forming along independent HZE particle tracks that will eventually be expressed in the form of chromosome exchanges of greater complexity. Chromosome exchanges result when radiation breaks a chromosome by severing the DNA of which it is composed. The cell has means to repair the chromosome and can splice the DNA together. On some occasions, if one chromosome break is close to another break, the cell might make a mistake and join the broken ends to inappropriate partners. Two or more breaks can be involved in the process; the greater the number of breaks, the more complicated the resulting chromosome exchange will be. We plan to test the hypothesis that these exchange complexity increases will be related to the primary particle energy and resultant maximum delta ray range. Higher particle energies will result in longer delta ray maximum ranges allowing more primary particle tracks to be close enough to irradiate a cell with delta electrons for any specific fluence (particle tracks per unit area; equivalent to dose). We plan to irradiate human skin cells with low and high energy beams matching LET (linear energy transfer; energy deposited per unit track length) and fluences so that the only variable will be particle energy. A dose response curve for chromosome exchanges irradiated with low particle fluences will be constructed. At the same time we will also look for breaks that fail to rejoin. One experiment will be conducted at low dose rate which will prevent damage from delta rays from being added to chromosome exchanges forming along primary particle tracks by spreading damage formation out in time and allowing some breaks to be repaired or exchanges to be completely formed before other breaks occurring later can be added to them. We also plan to grow populations from single surviving cells allowing us to measure the transmissibility of chromosome exchanges a number of cell generations removed from the irradiation. Chromosome aberrations are tightly linked to cancer development. With this study we hope to gain a better understanding of the processes underlying chromosome exchange formation following irradiation with HZE particles similar to those astronauts might encounter in space. We plan to determine the shape of the exchange dose response curve for these particles at low doses and at low dose rate, processes involving the misrepair of radiation induced damage that could lead to carcinogenesis. At the same time we will look at terminal deletions and incomplete exchanges resulting from the cell's failure to repair damage. These lesions are likely to kill the cell and eliminate it from a population that might develop into cancer. Finally, we will look for exchanges found only in a fraction of the cells making up a clonal population indicating genomic instability, the process that drives carcinogenesis. This project will endeavor to gain understanding on how delta ray interactions with primary particle tracks might affect these endpoints that are associated with processes that lead to cancer development and enable NASA to produce more accurate models for cancer risk estimates from exposure to galactic cosmic rays.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The overarching goal of this proposal is to determine the effect of a single charged particle track passing through a cell nucleus without any influence from d-rays contributed by independent tracks passing through the medium at the same time. This is the condition likely to occur in space and understanding the effects produced under these conditions is important in order to assess the dangers associated with galactic cosmic rays (GCR). Regrettably, simulating these conditions in the Earth environment is virtually impossible as even at the smallest doses, effects of a single track will be impacted by d-rays originating from independent tracks. It is the goal of this study to measure these track interaction effects on the chromosome exchange breakpoint dose response in order to separate them from those produced by single primary tracks. A thorough understanding of the mechanisms that underlie track interaction should allow the construction of more precise biophysical models of the effects of HZE particles on living organisms thereby reducing the uncertainty in cancer risk estimates NASA seeks in order to assess the hazards of GCR. In addition, track interaction is likely to have effects on particle beam therapy giving this study further societal and economic impact.
	SUPPLEMENTAL REPORTING FOR FINAL REPORT -- OCTOBER 2018 Our study involves the measurement of radiation induced chromosome aberrations produced by high energy heavy ions similar to those that are found in galactic cosmic rays (GCR). Chromosome aberrations are important radiation lesions that are associated with carcinogenesis. Chromosome aberrations are formed when radiation or some other treatment breaks chromosomes. Normally, the cell has the means to repair these breaks and joins the chromosomes back together. On occasion, if a number of breaks are close to one another, the cell can make a mistake and join the end one side of a break (break end) on one chromosome to one on another chromosome while the other break end from the first chromosome often is joined to the remaining break end on the second chromosome. This leads to an exchange of chromosomal segments that can be detected with chromosome analysis procedures. While many times, as in this example, exchanges involve only two breaks (termed simple exchanges), other times, three or more breaks may be involved in an exchange. For example, a break end from chromosome "A" can join to one from chromosome "B" while the other side of the break from "B" can join to one of the ends of a break on chromosome "C." The remaining break end from chromosome "C" can join to the open break end from chromosome "A" or it may join to a break end on a fourth chromosome and so on. When three or more breaks are involved, the exchange is described as a "complex exchange." Often the number and types of exchanges that can be observed are dependent on the type of radiation. Chromosome breaking damage produced by X-rays (the type of radiation used for most safety standards) are more or less located randomly within the cell nucleus. This allows exchanges be produced in two ways. In the first way, one beam of radiation (actually an electron that is moved by the radiation) produces all the damage necessary for the exchange to occur. This mechanism dominates when the dose (number of electrons) is low and the rate at which the electrons enter the cell is also low. Under these conditions the frequency of chromosome exchanges is proportional to the dose, that is, the higher the dose the more exchanges you would see. If you were to plot the data as a dot on a graph with the dose on the X-axis and the number of exchanges on the Y-axis you would see a straight line if you connected all the dots. The second way that is important at high doses and dose rates, where many more electron beams are present and some of the breaks that form the exchange can be formed by one electron beam while others are form by a second or even a third electron beam. When these exchanges are plotted as above a curved dose response results. The radiations in space are often very different. Instead of beams of electrons we find beams made from whole atoms

Task Progress:

that have lost their electrons (we call these ions). Many ions form when stars explode and these travel through space at speeds approaching the speed of light. These ions can't hurt us on Earth because they are deflected by the Earth's magnetic field but they could be harmful to astronauts traveling in deep space far beyond the Earth. Some of these ions like those that originate from iron atoms, produce most of their damage by the first mechanism we mentioned above (straight line) since all the damage they put into a cell is very close together and easily forms chromosome exchanges. But to make matters more complicated, they can also move electrons in the medium they are traveling in and these electrons (referred to as delta rays) might produce more damage that can make exchanges by the second mechanism (curved line).

Thankfully the doses in space are very small, so small that makes it difficult to simulate the dose and dose rates that we want to test in experiments on Earth. Yet we want good estimates of the effects of these ions so we can design spacecraft that will protect the astronauts from these potentially harmful radiation effects. The curved type effects are not likely to occur in space but are a real possibility for ground based simulations where we use large accelerators to speed up ions and use them to irradiate cells. If we have a good understanding of how many exchanges result from the curved mechanism in these experiments we can better estimate the damage produced by the single ions a cell is likely to encounter in space.

We expected the level of the curved portion of the response will depend on the types of ions used and the energy of those ions. The higher the energy of the primary particle (accelerated ion), the further the delta rays set in motion travel. We reasoned that the longer delta ray ranges from highest energy ions would have the best chance of causing a curved dose response. As the energy of the primary particle increased we would have expected to see more and more curvature in the dose response. This was the rationale of our project.

We used a series of ion beams of different energies that should have produced the same overall basic chemical damage in the cell (damage which can potentially break chromosomes) but whose geometrical distribution in the cell would have been quite different. If our understanding of these processes was correct, these different beams should have produced different levels of the curved response. As things turned out, with one exception, all the ion beams from very low energy lithium to extremely high energy iron produced essentially linear dose responses. The exception to these results was the calcium ion beam of intermediate energy. This caused us to rethink our hypothesis.

For curvature to be observed in the dose response, the delta rays from one primary particle would have to produce chromosome breaks that were close enough to be involved in an exchange with other breaks produced by a separate and independent primary particle, a process termed track interaction. We thought that the range of the delta rays would be of utmost importance in this process. As each of the different particle beams deposits the same amount of energy as it travels through the cell, those with shorter tracks deposit more energy along those tracks. Therefore, the lower energy particles produce a more concentrated distribution of energy induced damage closer to the primary particle track (path it takes through the cell), while the higher energy ions spread this same amount of damage over a longer distance. While the delta rays travel longer distances and can potentially encounter more independent tracks the concentration of that damage is less making it a more remote chance that a track interaction event could occur. The calcium ion beam is taken to represent an optimal energy where the delta rays have sufficiently long range to pass near independent tracks and produce damage with adequate density to allow track interaction events to occur with measurable frequency.

We have generated a vast sum of data. We plan to conduct a detailed analysis of all our results soon and present them in future publications.

OCTOBER 2017 REPORT:

Our study involves the measurement of radiation induced chromosome aberrations produced by high energy heavy ions similar to those that are found in galactic cosmic rays (GCR). Chromosome aberrations are important radiation lesions that are associated with carcinogenesis. Chromosome aberrations are formed when radiation or some other treatment breaks chromosomes. Normally, the cell has the means to repair these breaks and joins the chromosomes back together. On occasion, if a number of breaks are close to one another, the cell can make a mistake and join the end one side of a break (break end) on one chromosome to one on another chromosome while the other break end from the first chromosome often is joined to the remaining break end on the second chromosome. This leads to an exchange of chromosomal segments that can be detected with chromosome analysis procedures. While many times, as in this example, exchanges involve only two breaks (termed simple exchanges), other times, three or more breaks may be involved in an exchange. For example, a break end from chromosome "A" can join to one from chromosome "B" while the other side of the break from "B" can join to one of the ends of a break on chromosome "C." The remaining break end from chromosome "C" can join to the open break end from chromosome "A" or it may join to a break end on a fourth chromosome and so on. When three or more breaks are involved, the exchange is described as a "complex exchange."

Often the number and types of exchanges that can be observed are dependent on the type of radiation. Chromosome breaking damage produced by X-rays (the type of radiation used for most safety standards) are more or less located randomly within the cell nucleus. This allows exchanges be produced in two ways. In the first way, one beam of radiation (actually an electron that is moved by the radiation) produces all the damage necessary for the exchange to occur. This mechanism dominates when the dose (number of electrons) is low and the rate at which the electrons enter the cell is also low. Under these conditions the frequency of chromosome exchanges is proportional to the dose, that is, the higher the dose the more exchanges you would see. If you were to plot the data as a dot on a graph with the dose on the X-axis and the number of exchanges on the Y-axis you would see a straight line if you connected all the dots.

The second way that is important at high doses and dose rates, where many more electron beams are present and some of the breaks that form the exchange can be formed by one electron beam while others are form by a second or even a third electron beam. When these exchanges are plotted as above a curved dose response results.

The radiations in space are often very different. Instead of beams of electrons we find beams made from whole atoms that have lost their electrons (we call these ions). Many ions form when stars explode and these travel through space at speeds approaching the speed of light. These ions can't hurt us on Earth because they are deflected by the Earth's magnetic field but they could by harmful to astronauts traveling in deep space far beyond the Earth. Some of these ions like those that originate from iron atoms, produce most of their damage by the first mechanism we mentioned above (straight line) since all the damage they put in is real close together and easily forms chromosome exchanges. But to make matters more complicated, they can also move electrons and these electrons might produce more damage that can

make exchanges by the second mechanism (curved line).

Thankfully the doses in space are really small, so small that make it difficult to simulate the dose and dose rates that we want to test in experiments on Earth. Yet we want good estimates of the effects of these ions so we can design spacecraft that will protect the astronauts from these potentially harmful radiation effects. The curved type effects are not likely to occur in space but are a real possibility for ground based simulations where we use large accelerators to speed up ions and use them to irradiate cells. If we have a good understanding of how many exchanges result from the curved mechanism in these experiments we can better estimate the damage produced by the single ions a cell is likely to encounter in space.

We expect the level of the curved portion of the response will depend on the types of ions used and the energy of those ions. We will be using a series of ion beams of different energies that should produce the same overall basic chemical damage in the cell (damage which can potentially break chromosomes) but whose geometrical distribution in the cell will be quite different. If our understanding of these processes is correct, these different beams should produce different levels of the curved response. This is the rationale of our project.

Since our last report, we have been busy karyotyping (generating chromosome profiles) of the cells irradiated so far. These include fibroblasts (skin cells) and lymphocytes (blood cells) irradiated with iron, silicon, or calcium ions. We were also able to irradiate cells with iron ions at low dose rate this year. Lowering the dose rate should reduce the number of chromosome exchanges that are formed by the curved process and give us a better estimate of single ion effects.

We have recently published a paper with our collaborators concerning special statistical methods that might provide better procedures for analyzing our data.

Bibliography Type:	Description: (Last Updated: 10/29/2023)
Abstracts for Journals and Proceedings	Loucas BD, Cornforth MN. "The production of chromosomal exchanges by HZE ions of different energies but the same LET." 2018 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 22-25, 2018. 2018 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 22-25, 2018. , Jan-2018
Abstracts for Journals and Proceedings	Loucas BD, Cornforth MN. "The production of chromosomal exchanges by ions of different energies but the same LET: dose rate effects." Presented at the 63rd Annual Meeting of the Radiation Research Society, Grand Fiesta Americana Coral Beach, Cancun, Mexico, October 14-18, 2017. 63rd Annual Meeting of the Radiation Research Society, Grand Fiesta Americana Coral Beach, Cancun, Mexico, October 14-18, 2017. , Oct-2017
Abstracts for Journals and Proceedings	Loucas BD, Cornforth MN. "The production of simple and complex chromosome exchanges by HZE ions of different energies but the same LET." Presented at the 64th Annual Meeting of the Radiation Research Society, Chicago, IL, September 23-26, 2018. 64th Annual Meeting of the Radiation Research Society, Chicago, IL, September 23-26, 2018. , Sep-2018
Articles in Peer-reviewed Journals	Cornforth MN, Durante M. "Radiation quality and intra-chromosomal aberrations: Size matters." Mutat Res. 2018 Dec;836(Pt A):28-35. Review. Epub 2018 May 5. https://doi.org/10.1016/j.mrgentox.2018.05.002 ; PubMed PMID: 30389158 , Dec-2018
Articles in Peer-reviewed Journals	Shuryak I, Loucas BD, Cornforth MN. "Seeking beta: experimental considerations and theoretical implications regarding the detection of curvature in dose-response relationships for chromosome aberrations." Radiat Res. 2017 Jan;187(1):7-19. https://doi.org/10.1667/RR14520.1 ; PubMed PMID: 28085640 , Jan-2017
Articles in Peer-reviewed Journals	Cornforth M, Shuryak I, Loucas B. "Lethal and nonlethal chromosome aberrations by gamma rays and heavy ions: a cytogenetic perspective on dose fractionation in hadron radiotherapy." Transl Can Res. 2017 Jul;6 Suppl 5:S769-S778. http://dx.doi.org/10.21037/tcr.2017.05.16 , Jul-2017
Articles in Peer-reviewed Journals	Shuryak I, Loucas BD, Cornforth MN. "Straightening beta: Overdispersion of lethal chromosome aberrations following radiotherapeutic doses leads to terminal linearity in the alpha-beta model." Front Oncol. 2017 Dec 21;7:318. https://doi.org/10.3389/fonc.2017.00318 ; PubMed PMID: 29312888; PubMed Central PMCID: PMC5742594 , Dec-2017